

Earl Chirchir

Graduate Mechatronic Engineer (MSc Distinction) • Tailored for Graduate Mechatronic Engineer at Recruiters & Engineering Teams

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Work Authorization: Graduate Visa (Full Time Right to Work)

PROFESSIONAL SUMMARY

Mechatronic Engineer (MSc, Distinction) with 3+ years experience in the design, development and integration of automation and embedded systems within industrial and manufacturing environments. Strong technical capability in robotics and trajectory planning, control systems, embedded programming (C++/Python) and CAD/CAM design, with proven experience delivering solutions from concept through implementation.

CORE SKILLS & TECHNOLOGIES

- **Robotics & Automation:** Cobot Trajectory Planning (MATLAB, RoboDK), ABB CRB 15000, Bang-Coast-Bang Time Law
- **CAD/CAM Design:** NX Siemens, Autodesk Inventor, SolidWorks
- **Embedded & Programming:** C++ (Arduino/ESP32), Python, MATLAB, Electronic Sensors & Actuators
- **Circuit & PCB Design:** KiCAD, AutoDesk Eagle, Fritzing, Proteus
- **Industrial Systems & Data:** Industrial HMIs (Nextion), SQL Databases, Power BI Analytics, Predictive Diagnostics

WORK EXPERIENCE

Development Engineer — OJE Engineering (Kenya) | 11/2022 – 12/2024

- Led maintenance of an IoT data logging telemetry system for tea-picking machinery, improving data transmission efficiency.
- Designed and implemented Nextion HMI displays for industrial weighing bins, increasing machine operational uptime.
- Supervised configuration of remote web servers and SQL databases using Postman for secure IoT data handling.
- Programmed embedded systems (Arduino Nano) for industrial automation and JSON data integration with Power BI.

Enforcement & Consumer Protection Engineer — EPRA (Kenya) | 01/2019 – 12/2019

- Supported regulatory operations across petroleum and electric power sectors, reviewing electricity meters for accuracy.

EDUCATION

MSc Mechatronic Systems (Pass with Distinction) — Kingston University, UK | 01/2024 – 02/2025

Dissertation: Trajectory Planning for ABB CRB 15000 Robot Using Bang-Coast-Bang Time Law (Grade 81%).

Key Modules: Control Systems (79%), Engineering Research Techniques (85%), Mechatronic Design.

BSc Mechatronic Engineering (Upper Second Class Honors 2:1) — JKUAT | 09/2013 – 11/2018